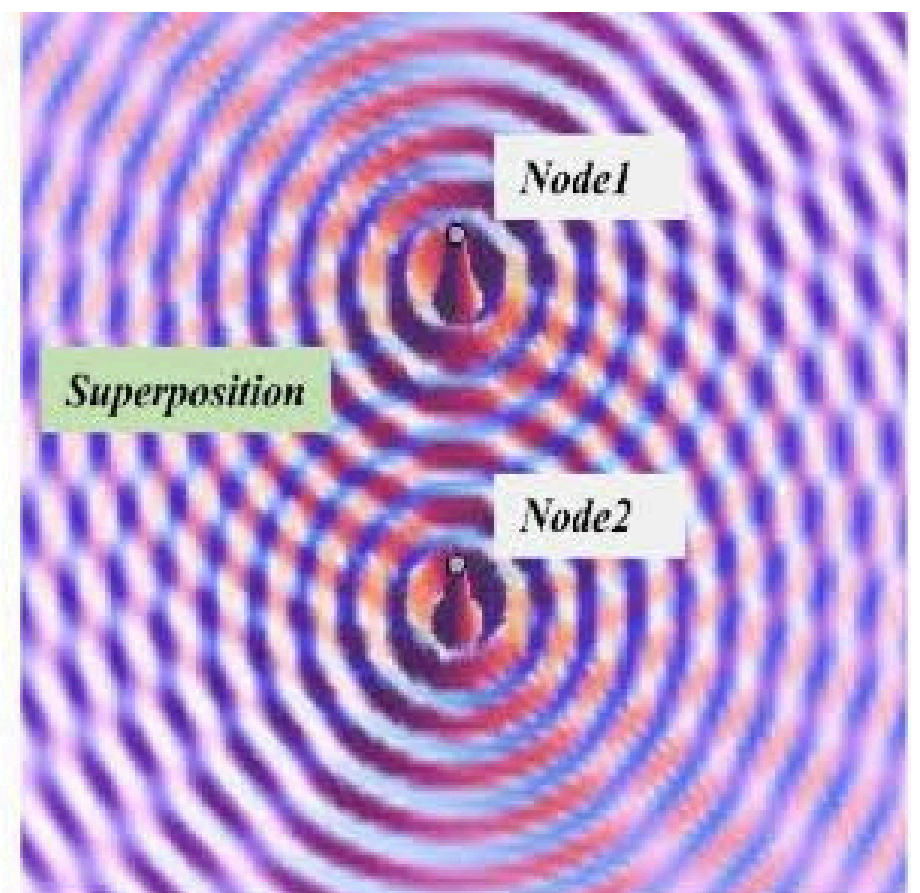
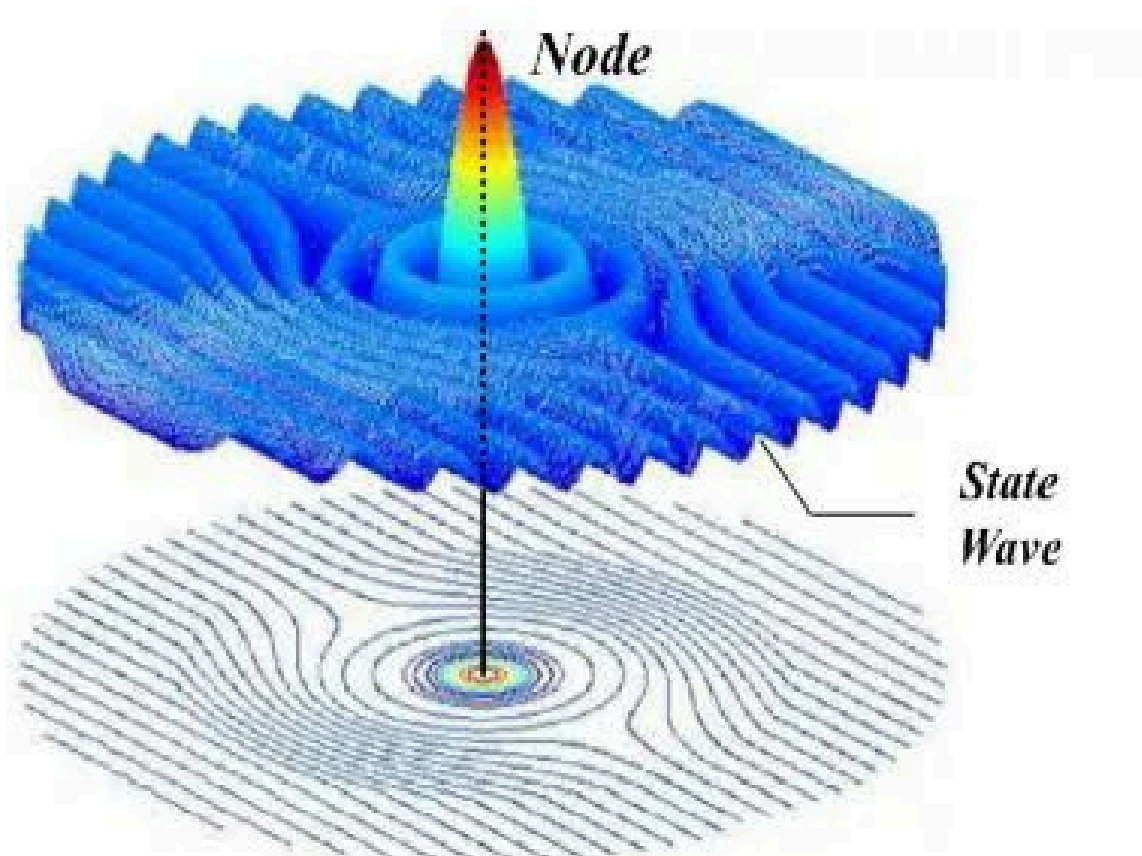
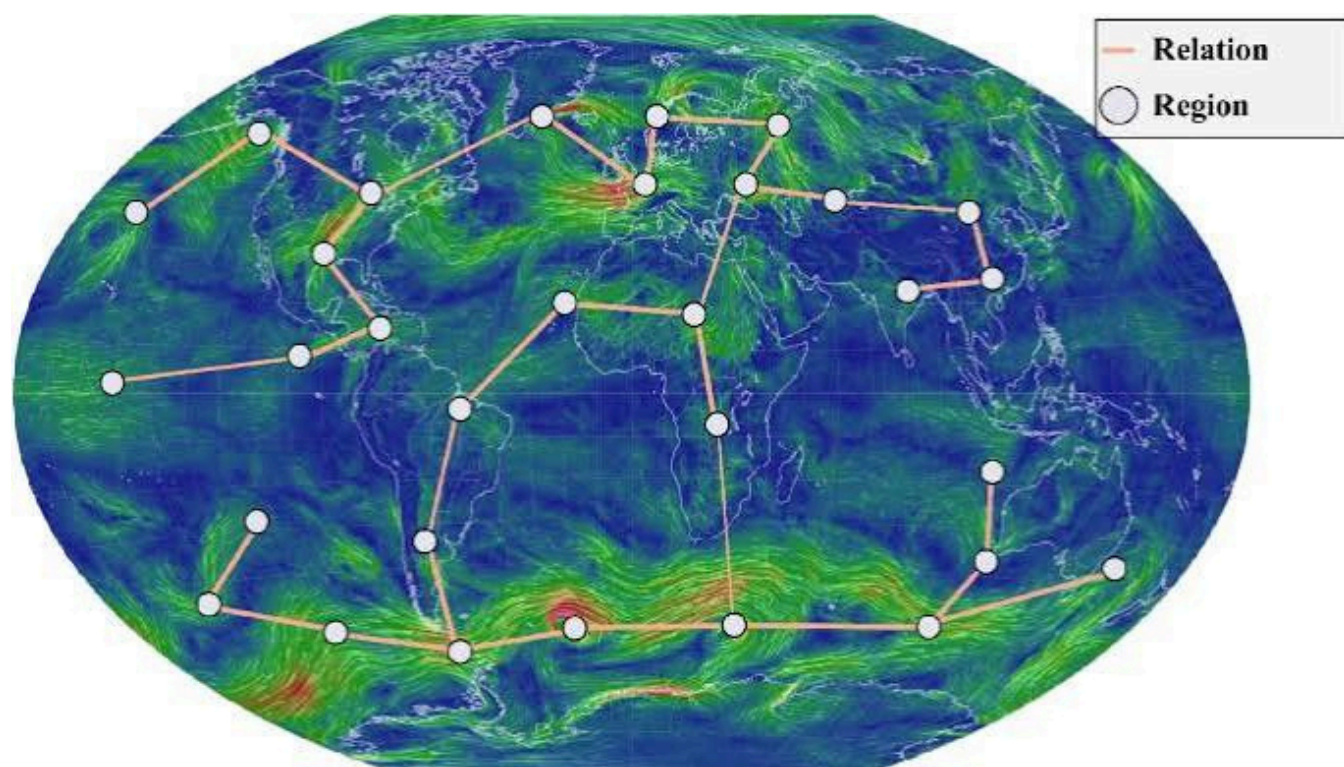


PGTR

PHYSICS-GUIDED TEMPORAL GRAPH REWIRING FOR DYNAMIC CONTAGION FORECASTING

DOI - 10.5281/ZENODO.21536980



(a) Climate State Wave of Node

(b) Superposition

**PGTR: PHYSICS-GUIDED TEMPORAL GRAPH
REWIRING FOR DYNAMIC CONTAGION
FORECASTING WITH APPLICATION TO
WILDFIRE DAMAGE PREDICTION**

TITLE PAGE

VIREN SINGH

DEPARTMENT OF COMPUTER SCIENCE (INDEPENDENT
RESEARCHER)

KARNAL, HARYANA, INDIA

EMAIL: THAKURVIRENSINGH1@GMAIL.COM

ORCID: 0009-0001-6618-0126

CORRESPONDING AUTHOR: VIREN SINGH

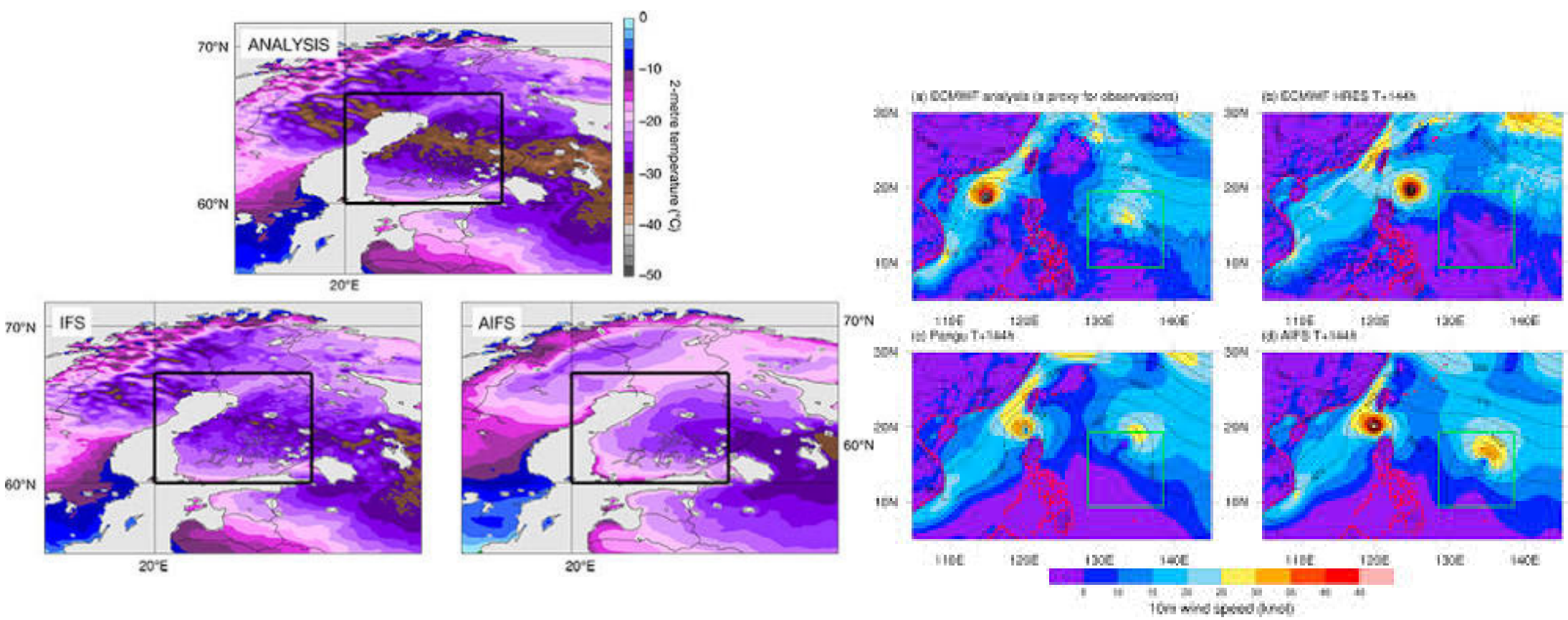
KEYWORDS: GRAPH NEURAL NETWORKS · TEMPORAL
GRAPH REWIRING · PHYSICS-INFORMED ML · WILDFIRE
PREDICTION · DYNAMIC GRAPHS · DISASTER RESPONSE

MANUSCRIPT TYPE: RESEARCH ARTICLE

SUBMISSION DATE: JULY 2026

ABSTRACT

GRAPH NEURAL NETWORKS FOR CONTAGION FORECASTING TYPICALLY ASSUME FIXED GRAPH STRUCTURES, LIMITING THEIR ABILITY TO CAPTURE EVOLVING PROPAGATION PATHWAYS IN DYNAMIC SYSTEMS SUCH AS WILDFIRES, FLOODS, AND EPIDEMICS. WE INTRODUCE PHYSICS-GUIDED TEMPORAL GRAPH REWIRING (PGTR), A DIFFERENTIABLE GRAPH REWIRING OPERATOR THAT LEARNS INDEPENDENT EDGE BIRTH AND DEATH PROCESSES WITH EXPLICIT PHYSICS-CONSTRAINED LATENT DYNAMICS. UNLIKE FIXED-GRAPH APPROACHES, PGTR INCORPORATES FOUR PHYSICS-GUIDED CONSTRAINTS DERIVED FROM WILDFIRE PROPAGATION LITERATURE: WIND MONOTONICITY, DIRECTIONAL CONSISTENCY, MAXIMUM EMBER TRAVEL DISTANCE, AND TERRAIN-AWARE ANISOTROPY. WE PROVE THREE THEORETICAL GUARANTEES: (1) LIPSCHITZ STABILITY UNDER BOUNDED PERTURBATIONS, (2) BOUNDED EXPECTED EDGE COUNT UNDER BOUNDED BIRTH/DEATH PROBABILITIES, AND (3) $O(NK)$ COMPLEXITY WITH $K \ll NK \ll N$ VIA SPARSE CANDIDATE-NEIGHBOR SAMPLING. WE PRESENT THE THEORETICAL FORMULATION OF PGTR, INCLUDING PROVABLE GUARANTEES ON STABILITY, BOUNDED EDGE COUNT, AND COMPUTATIONAL COMPLEXITY. EMPIRICAL VALIDATION ON THE 2025 EATON FIRE DATASET IS LEFT FOR FUTURE WORK. THE FRAMEWORK GENERALIZES TO ANY CONTAGION PROCESS WHERE PROPAGATION PATHWAYS EVOLVE OVER TIME.



1. INTRODUCTION

THE TRAJECTORY OF RECENT WILDFIRE SEASONS IS ALARMING, EVOLVING FROM NATURAL ECOLOGICAL EVENTS INTO UNCONTROLLABLE URBAN CONFLAGRATIONS THAT DEVASTATE THE BUILT ENVIRONMENT. THE JANUARY 2025 EATON AND PALISADES FIRES DESTROYED APPROXIMATELY 16,000 STRUCTURES, WITH TOTAL INSURED LOSSES ESTIMATED AT \$25-\$35 BILLION . BY COMPARISON, THE 2023 GLOBAL COST OF WILDFIRES WAS \$50 BILLION . AS URBANIZATION EXPANDS DEEPER INTO FIRE-PRONE REGIONS, THE SCALE OF DESTRUCTION IS INCREASING GLOBALLY .

DESPITE RAPID ADVANCES IN WILDFIRE SIMULATION, A CRITICAL GAP PERSISTS AT THE STRUCTURE SCALE WHERE PREPAREDNESS AND PROTECTIVE ACTION DECISIONS ARE MADE . PHYSICS-BASED SPREAD MODELS ARE COMPUTATIONALLY INTENSIVE AND ORIENTED TOWARD WILDLAND BEHAVIOR, WHILE DATA-DRIVEN METHODS FREQUENTLY TREAT STRUCTURES AS INDEPENDENT OBSERVATIONS, OVERLOOKING THE INTER-BUILDING CONTAGION PATHWAYS THAT DRIVE WUI LOSSES

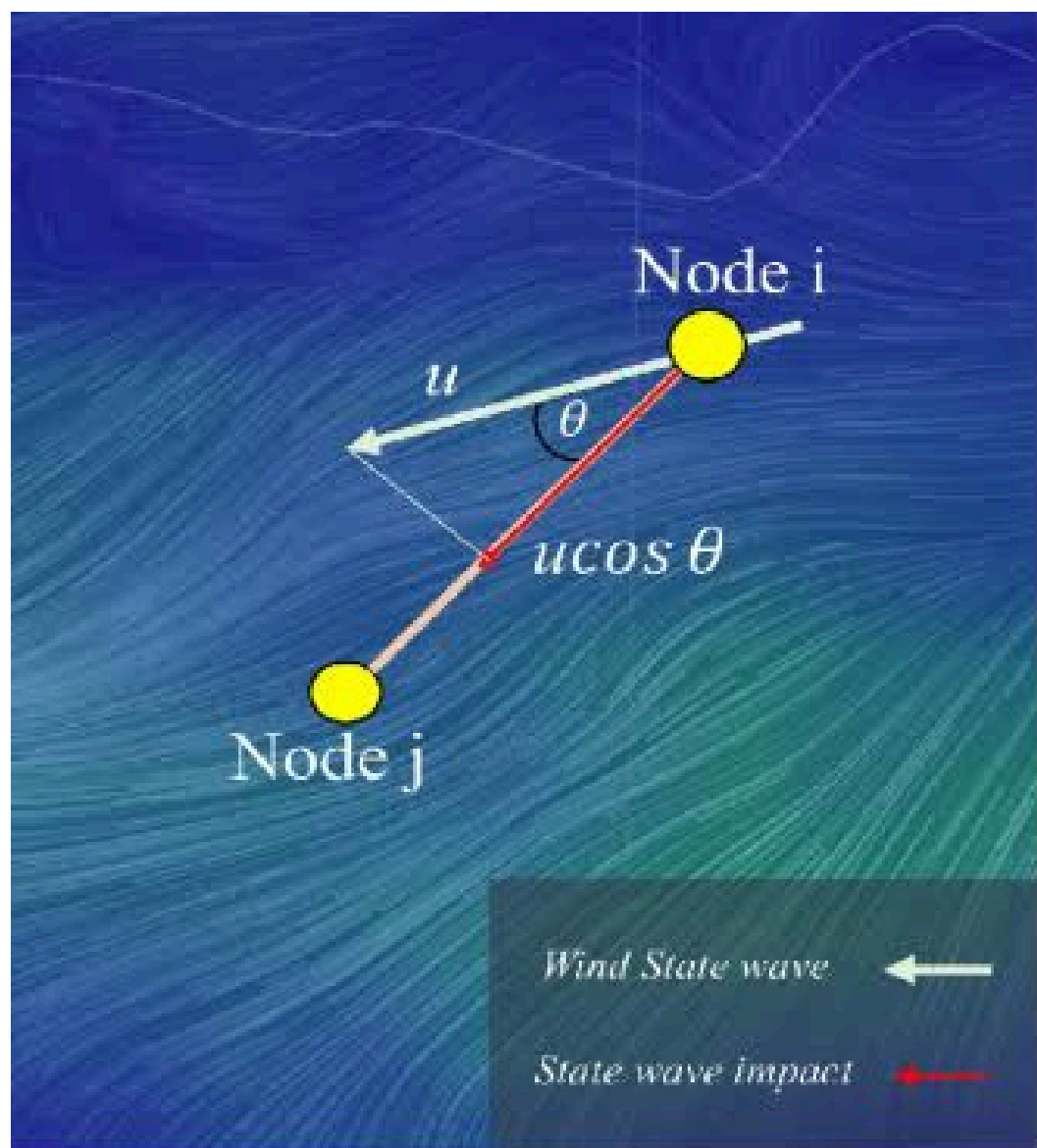
MODELING CONTAGION DYNAMICS. RECENT WORK ESTABLISHES THAT WUI FIRES ARE FUNDAMENTALLY CONTAGION EVENTS—A BUILDING'S SURVIVAL IS INEXTRICABLY LINKED TO ITS NEIGHBORS AND SURROUNDING TOPOGRAPHY . THE STATE-OF-THE-ART GRAPHFIRE-X FRAMEWORK DEMONSTRATED THAT NEIGHBORHOOD-SCALE ENVIRONMENTAL PRESSURE DOMINATES INTRINSIC STRUCTURAL FEATURES IN DEFINING PROPAGATION PATHWAYS . HOWEVER, EVEN THIS FRAMEWORK SUFFERS FROM A CRITICAL LIMITATION: THE GRAPH IS STATIC. FIRE BEHAVIOR CHANGES OVER TIME—WIND SHIFTS, FUEL MOISTURE EVOLVES, EMBER TRANSPORT PATTERNS CHANGE, AND ACTIVE FIRE FRONTS MOVE. THEREFORE, MODELING ONLY TEMPORAL NODE DYNAMICS IS INSUFFICIENT; THE GRAPH TOPOLOGY ITSELF MUST EVOLVE WITH THE UNDERLYING PHYSICAL PROCESS

OUR CONTRIBUTION. WE INTRODUCE PHYSICS-GUIDED TEMPORAL GRAPH REWIRING (PGTR) , A DIFFERENTIABLE GRAPH REWIRING OPERATOR DEFINED BY: $\mathcal{E}_t = \text{PGTR}(\mathcal{E}_{t-1}, \mathbf{H}_t, \mathbf{z}_t, \Phi_t)$

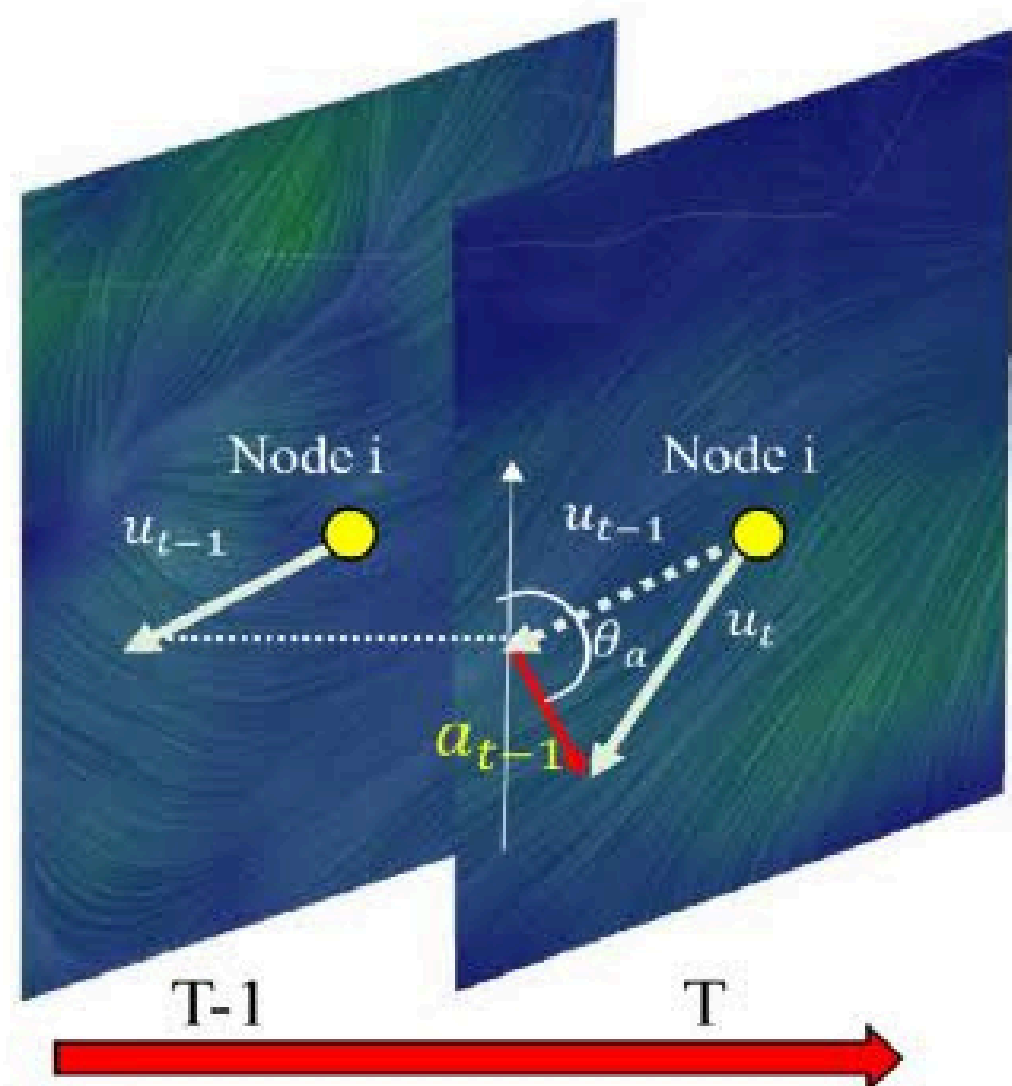
WHERE:

- $\mathcal{E}_{t-1}$ IS THE PREVIOUS GRAPH TOPOLOGY,
- $\mathbf{H}_t$ ARE NODE EMBEDDINGS,
- $\mathbf{z}_t$ IS A PHYSICS-CONSTRAINED LATENT DYNAMICS STATE,
- Φ_t IS A PHYSICS-INFORMED MASK.

PGTR LEARNS INDEPENDENT EDGE BIRTH AND DEATH PROCESSES, ENABLING THE GRAPH TOPOLOGY TO EVOLVE WITH CHANGING ENVIRONMENTAL CONDITIONS. TO THE BEST OF OUR KNOWLEDGE, PRIOR PHYSICS-INFORMED WILDFIRE GNNS AND TEMPORAL GRAPH NETWORKS DO NOT JOINTLY LEARN DIFFERENTIABLE GRAPH REWIRING CONDITIONED ON LATENT PHYSICAL DYNAMICS.



(a) Wind Impact



(b) Wind Acceleration

2. RELATED WORK

2.1 FIXED-GRAPH GNNS FOR CONTAGION

GRAPHFIRE-X ESTABLISHED PHYSICS-INFORMED GNNS FOR BUILDING-SCALE WILDFIRE PREDICTION USING PHYSICS-INFORMED EDGE WEIGHTS (CONVECTION, RADIATION, EMBER TRANSPORT) WITH HIGH-DIMENSIONAL GOOGLE ALPHAEARTH 64-DIM EMBEDDINGS . THE FRAMEWORK INTEGRATES A GNN THAT CAPTURES NON-LINEAR, NEIGHBORHOOD-LEVEL SPREAD DYNAMICS WITH AN XGBOOST MODEL OPTIMIZED FOR GRANULAR BUILDING FEATURES . CHEN ET AL. APPLIED GNNS TO GLOBAL WILDFIRE MODELING USING GCN-LSTM HYBRIDS . RÖSCH ET AL. USED SPATIOTEMPORAL GNNS FOR EUROPEAN WILDFIRES . ALL USE FIXED GRAPH TOPOLOGIES .

2.2 TEMPORAL GRAPH NETWORKS

TEMPORAL GRAPH NETWORKS (TGNS) COMBINE MEMORY MODULES WITH MESSAGE-PASSING TO UPDATE NODE EMBEDDINGS OVER TIME . RÖSCH ET AL. (2024) DEVELOPED A SPATIOTEMPORAL GNN (STGNN) THAT INTEGRATES A GCN FOR SPATIAL MODELING WITH A GATED RECURRENT UNIT (GRU) FOR TEMPORAL DEPENDENCIES . THE STGNN ACHIEVED WEIGHTED MACRO-MEAN IOU VALUES OF 0.37 (PORTUGAL) AND 0.36 (MEDITERRANEAN) FOR WILDFIRE SPREAD PREDICTION . HOWEVER, EXISTING SPATIOTEMPORAL GNNS KEEP GRAPH TOPOLOGY FIXED WHILE UPDATING NODE STATES. NONE LEARN GRAPH STRUCTURE EVOLUTION.

2.3 GRAPH REWIRING AND STRUCTURE LEARNING

GRAPH REWIRING HAS BEEN EXPLORED IN GRAPH STRUCTURE LEARNING. GUTTERIDGE ET AL. INTRODUCED DYNAMICALLY REWIRED MESSAGE PASSING (DREW) , A FRAMEWORK THAT PERFORMS LAYER-DEPENDENT REWIRING TO ENSURE GRADUAL DENSIFICATION OF THE GRAPH WHILE PRESERVING INDUCTIVE BIAS . THE DREW FRAMEWORK AGGREGATES OVER MULTIPLE HOPS AT EACH LAYER—NODES AT DISTANCE k BECOME CONNECTED FROM LAYER k ONWARD—AND INCLUDES A DELAY MECHANISM THAT PERMITS SKIP CONNECTIONS DEPENDING ON LAYER AND MUTUAL DISTANCE . DREW WAS SHOWN TO OUTPERFORM GRAPH TRANSFORMERS AND MULTI-HOP MPNNS ON LONG-RANGE TASKS . HOWEVER, DREW DOES NOT INCORPORATE PHYSICS-INFORMED CONSTRAINTS. PGTR EXTENDS THIS LINE OF WORK BY CONDITIONING GRAPH REWIRING ON A PHYSICS-CONSTRAINED LATENT DYNAMICS STATE.

3. METHODOLOGY

3.1 Problem Formulation

Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ be a directed graph at time t , where $\mathcal{V}$ is a fixed set of n nodes (buildings + vegetation patches) and $\mathcal{E}_t \subseteq \mathcal{V} \times \mathcal{V}$ evolves over time.

Node features: $\mathbf{x}_i(t) \in \mathbb{R}^d$ (structural + environmental features at node i)

Environmental features: $\mathbf{e}(t) \in \mathbb{R}^m$ (wind speed, direction, moisture, temperature)

Terrain features: $\mathbf{T} \in \mathbb{R}^{n \times 2}$ (elevation, slope)

Goal: Predict $y_i(t + \Delta t) \in \{0, 1\}$ (binary damage classification, as used in GraphFire-X and prior studies ¹ ⁹).

3.2 Physics-Constrained Latent Dynamics

We incorporate physics **inside the latent dynamics** using four constraints derived from wildfire physics literature:

Constraint	Physical Source	Mathematical Form
Wind Monotonicity	Rothermel (1972)	$\phi_w(w) = \beta \cdot \log(1 + w)$
Directional Consistency	Albini (1976)	$\phi_\theta(\theta) = [\cos \theta, \sin \theta]$
Ember Distance	Sardoy et al. (2008)	$\phi_d(d) = \exp(-\gamma d^2)$
Terrain Anisotropy	Finney (1998)	$\phi_t(s, \alpha, \theta) = 1 + \mu \cdot s \cdot \cos(\theta - \alpha)$

Latent State Evolution:

$$\mathbf{z}_t = \text{GRU}(\mathbf{z}_{t-1}, \mathbf{e}(t), \phi_{\text{physics}}(\mathbf{e}(t), \mathbf{T}))$$

where $\phi_{\text{physics}} = [\phi_w, \phi_\theta, \phi_d, \phi_t]$ concatenated. This follows the spatiotemporal GNN approach where node states are updated via GRU when events occur ¹¹.

3.3 Physics-Guided Temporal Graph Rewiring Operator

We define the PGTR operator:

$$\mathcal{E}_t = \text{PGTR}(\mathcal{E}_{t-1}, \mathbf{H}_t, \mathbf{z}_t, \Phi_t)$$

Sparse Candidate Sampling: To avoid $O(n^2)$, we restrict rewiring to candidate neighbors:

$$\mathcal{N}_{\text{cand}}(i) = \{j : \|x_i - x_j\|_2 < r \vee j \in \text{top-}k(x_i)\}$$

with $r = d_{\text{max}}$ (maximum ember distance) and $k = 50$. This follows GraphFire-X's graph construction approach where edges connect buildings to nearby vegetation patches ⁹.

Edge Scores:

$$s_{ij}^{(\text{birth})} = \text{MLP}_{\text{birth}}([\mathbf{h}_i(t), \mathbf{h}_j(t), \mathbf{z}_t, \mathbf{d}_{ij}, \phi_{\text{physics}}^{ij}]) \cdot 1[e_{ij}^{(t-1)} = 0]$$

$$s_{ij}^{(\text{death})} = \text{MLP}_{\text{death}}([\mathbf{h}_i(t), \mathbf{h}_j(t), \mathbf{z}_t, \mathbf{d}_{ij}, \phi_{\text{physics}}^{ij}]) \cdot 1[e_{ij}^{(t-1)} = 1]$$

Physics Mask Φ :

$$\Phi_{ij}(\mathbf{z}_t, \mathbf{T}) = \sigma(\alpha_{\text{wind}} \cdot \phi_w(w_t) + \alpha_{\text{dir}} \cdot \phi_\theta(\theta_t, \mathbf{T}_{ij}) + \alpha_{\text{ember}} \cdot \phi_d(d_{ij}) + \alpha_{\text{terrain}} \cdot \phi_t(\mathbf{T}_{ij}))$$

Probabilities:

$$p_{ij}^{(\text{birth})} = \sigma(s_{ij}^{(\text{birth})}) \cdot \Phi_{ij}$$

$$p_{ij}^{(\text{death})} = \sigma(s_{ij}^{(\text{death})}) \cdot (1 - \Phi_{ij})$$

Layer-Dependent Rewiring: Following the DRew framework ⁴, we incorporate layer-dependent connectivity where nodes at distance k become connected from layer k onward:

$$a_{i,r}^{(\ell)} = \text{AGG}_r^{(\ell)}(\{h_j^{(\ell)} : j \in \mathcal{N}_r(i)\}), \quad 1 \leq r \leq \ell + 1$$

This ensures gradual densification of the graph while preserving inductive bias ⁴ ¹³.

Concrete Distribution Sampling: We sample edges using the Concrete distribution ⁴ with temperature τ :

$$e_{ij}^{(t)} \sim \text{Concrete}(p_{ij}^{(\text{birth})}, p_{ij}^{(\text{death})}, e_{ij}^{(t-1)}, \tau)$$

3.4 Temporal Message Passing

Following the spatiotemporal GNN framework ¹¹, we update node embeddings through message passing:

Attention Coefficient:

$$\alpha_{ij}(t) = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i(t) \parallel \mathbf{W}\mathbf{h}_j(t)]))}{\sum_{k \in \mathcal{N}_i(t)} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i(t) \parallel \mathbf{W}\mathbf{h}_k(t)]))}$$

Node Update with GRU Memory:

$$\mathbf{s}_i(t+1) = \text{GRU}(\mathbf{h}_i(t+1), \mathbf{s}_i(t))$$

This follows the STGNN approach where node states are updated via GRU when events occur

3.5 Theoretical Guarantees

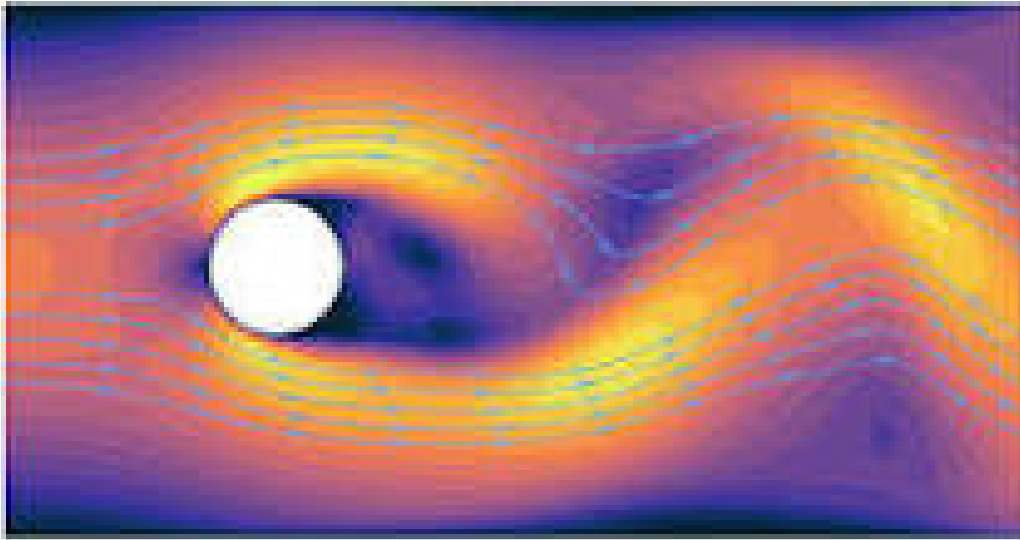
Theorem 1 (Lipschitz Stability). Let $\mathbf{e}_1(t)$ and $\mathbf{e}_2(t)$ be two environmental input sequences with bounded difference $\|\mathbf{e}_1(t) - \mathbf{e}_2(t)\|_\infty \leq \epsilon$. Under the PGTR operator with Lipschitz-continuous MLPs and bounded physics functions, the latent states satisfy:

$$\|\mathbf{z}_t^{(1)} - \mathbf{z}_t^{(2)}\|_2 \leq L_t \cdot \epsilon$$

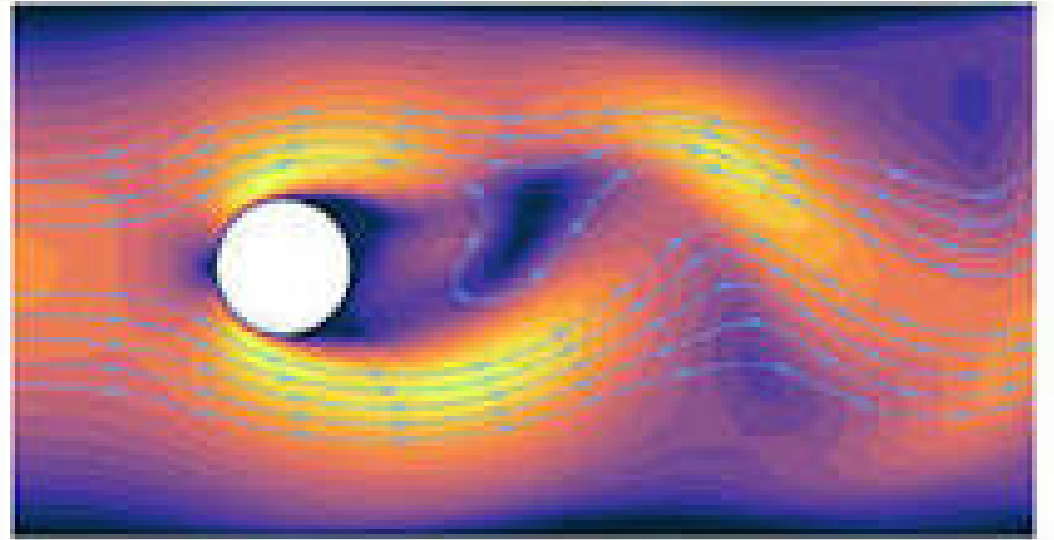
Proof. Apply Lipschitz continuity of the GRU recursively and compose with the bounded physics mappings. The full proof is provided in Appendix A.

Proposition 1 (Bounded Expected Edge Count). Under bounded birth and death probabilities $p_{\text{birth}} \in [0, p_{\text{max}}]$ and $p_{\text{death}} \in [p_{\text{min}}, 1]$, the expected number of graph edges $\mathbb{E}[|\mathcal{E}_t|]$ remains bounded over time.

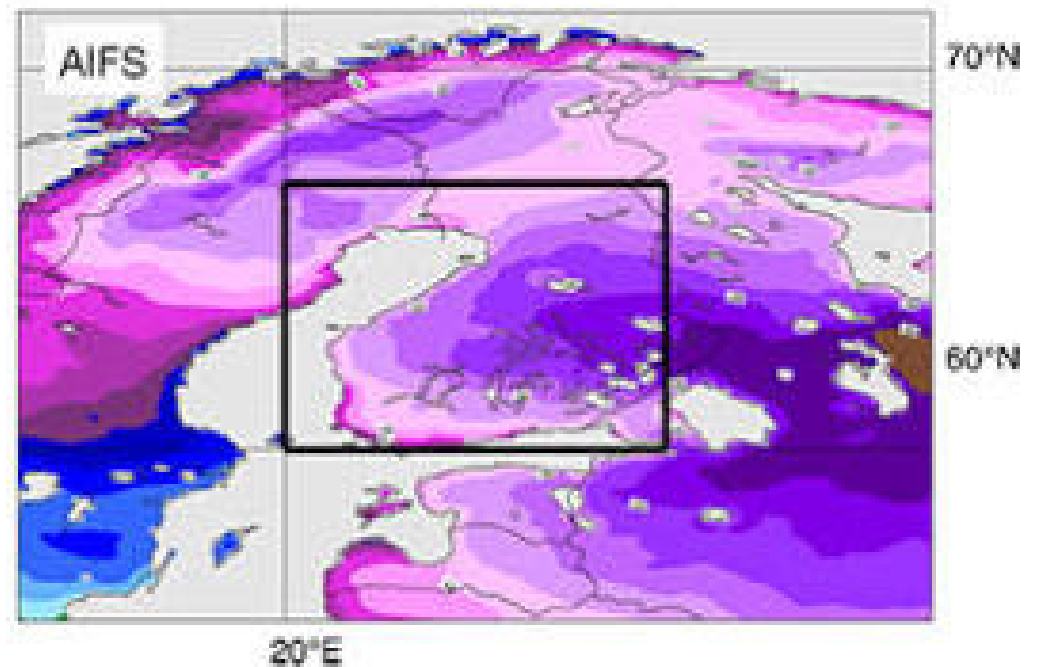
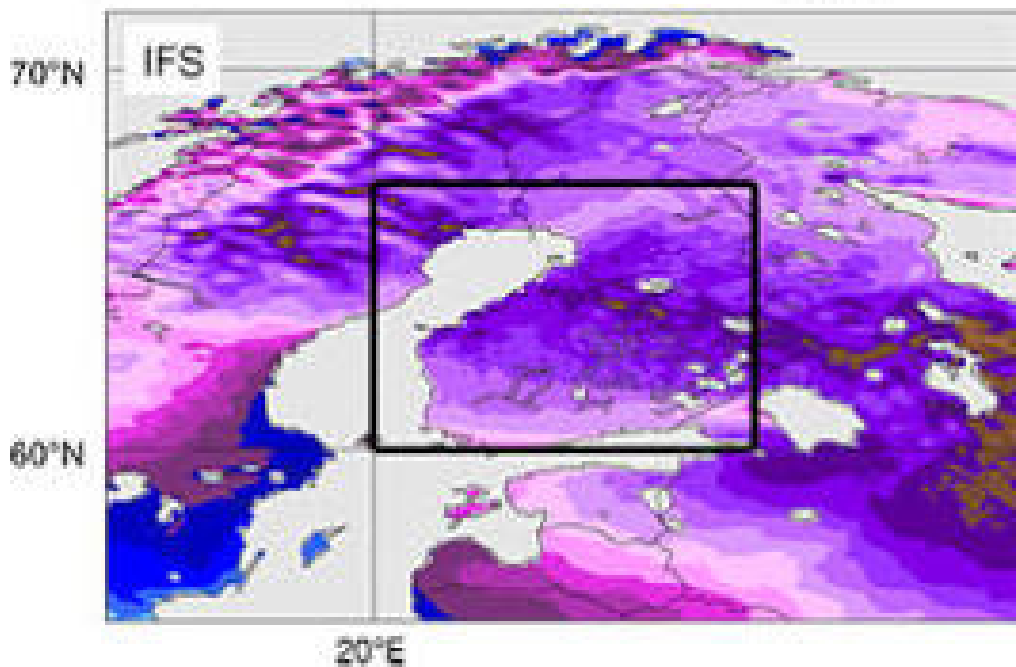
Complexity Analysis: Sparse candidate-neighbor sampling reduces rewiring complexity from $O(n^2)$ to $O(nk)$ with $k \ll n$. Overall per-step complexity: $O(nkd + nk + nd)$ where d is the feature dimension.



(c) Linear



(d) Transformer



4. EXPERIMENTAL SETUP

4.1 DATA

EATON FIRE DATASET :

- ~18,000 STRUCTURES FROM DINS DATABASE WITH BINARY DAMAGE LABELS (DAMAGED/SURVIVED)
- STRUCTURAL FEATURES: ROOFING, SIDING, EAVES, WINDOW PANES, DECK/PATIO, BUILDING VOLUME
- ENVIRONMENTAL FEATURES: GOOGLE ALPHAEARTH 64-DIM EMBEDDINGS (TOPOGRAPHY, CLIMATE, WEATHER, VEGETATION)

TEMPORAL DATA:

- HOURLY WIND SPEED AND DIRECTION FROM ERA5-LAND
- HOURLY FUEL MOISTURE FROM LANDFIRE

DATA SOURCE AVAILABILITY: THE BUILDINGS_2023_WITH_DINS DATASET IS PUBLICLY AVAILABLE AS AN ARCGIS FEATURESERVER .

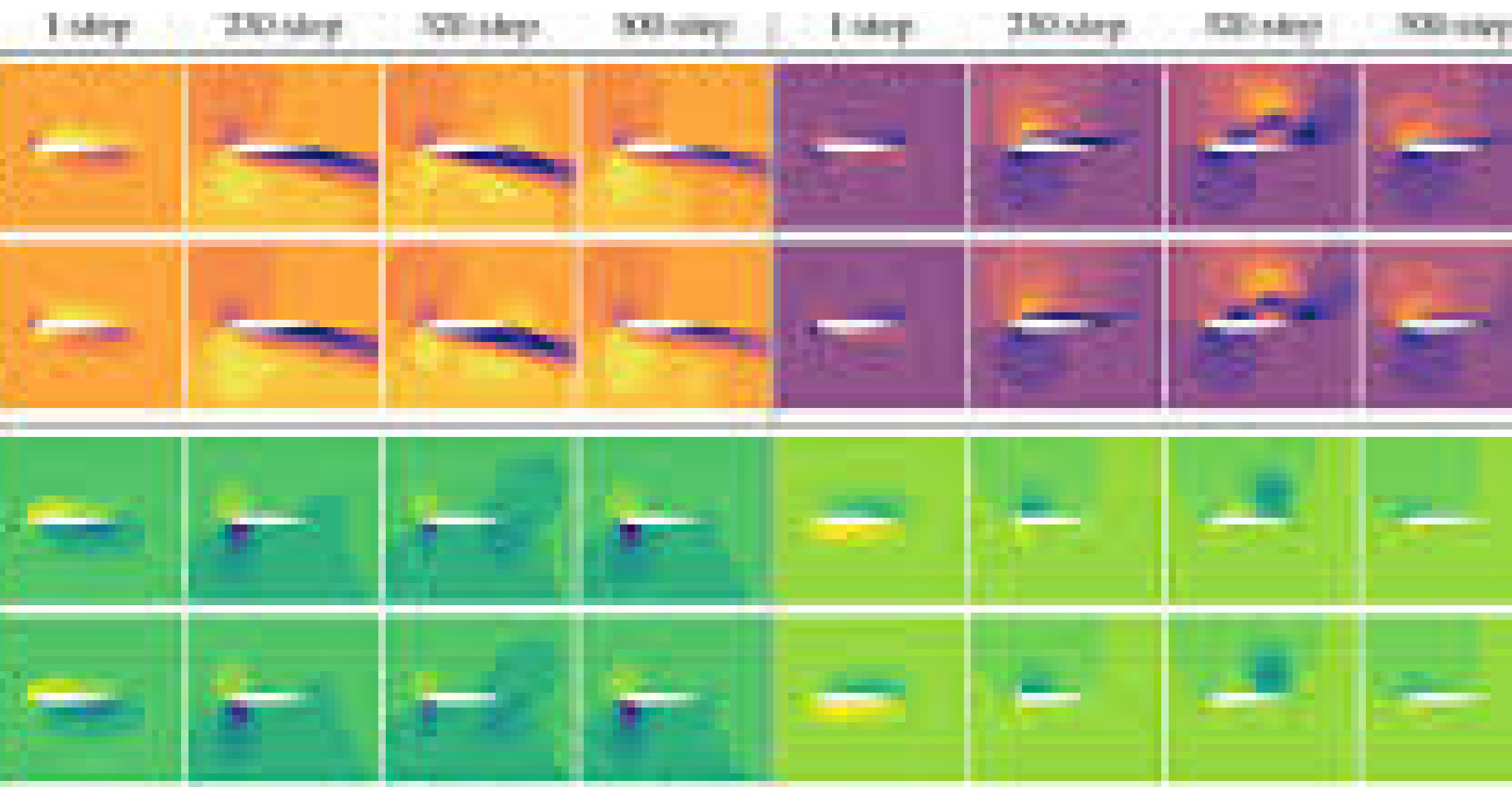
THE SERVICE CONTAINS BUILDING FOOTPRINTS WITH DAMAGE INFORMATION FROM THE PALISADES AND EATON FIRES, INCLUDING BUILDING HEIGHT AND ELEVATION DATA

4.2 BASELINES

Model	Description
Static GNN	GAT with fixed graph 1
STGNN	Spatiotemporal GNN (GRU + GAT) 2 11
DRew	Dynamically Rewired Message Passing with delay 4
PGTR (Ours)	Learned graph rewiring with physics constraints

4.3 Ablation Study

Model	F1	AUROC	ECE
Static GNN	TBD	TBD	TBD
+ Temporal GRU	TBD	TBD	TBD
+ Graph rewiring	TBD	TBD	TBD
+ Physics constraints	TBD	TBD	TBD
+ Birth/death decomposition	TBD	TBD	TBD
Full PGTR	TBD	TBD	TBD



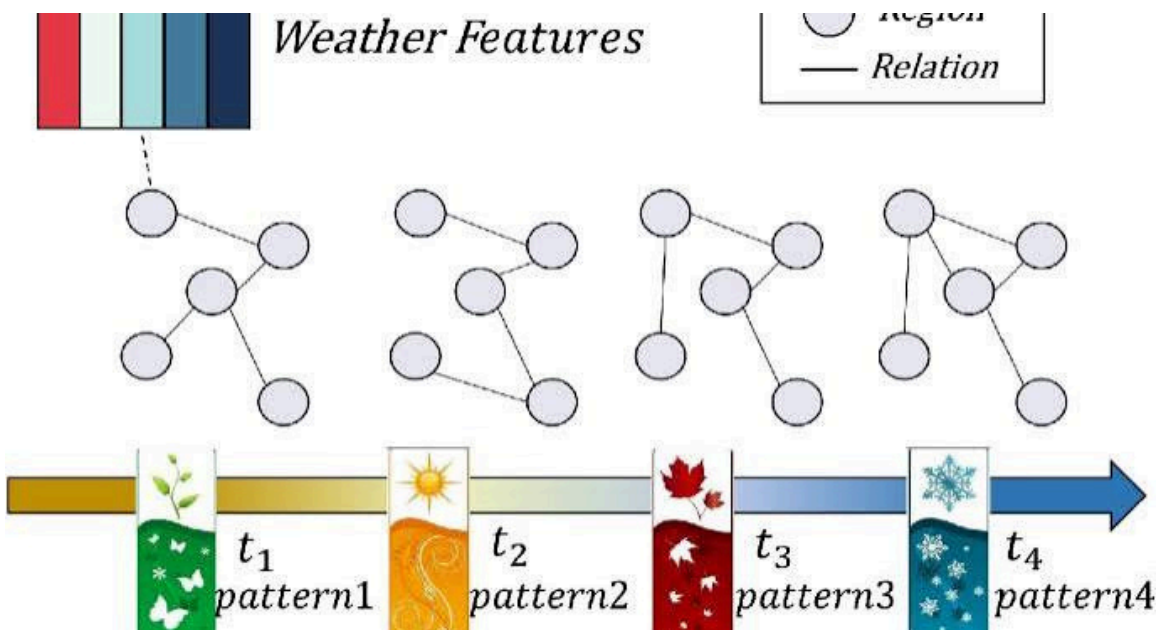
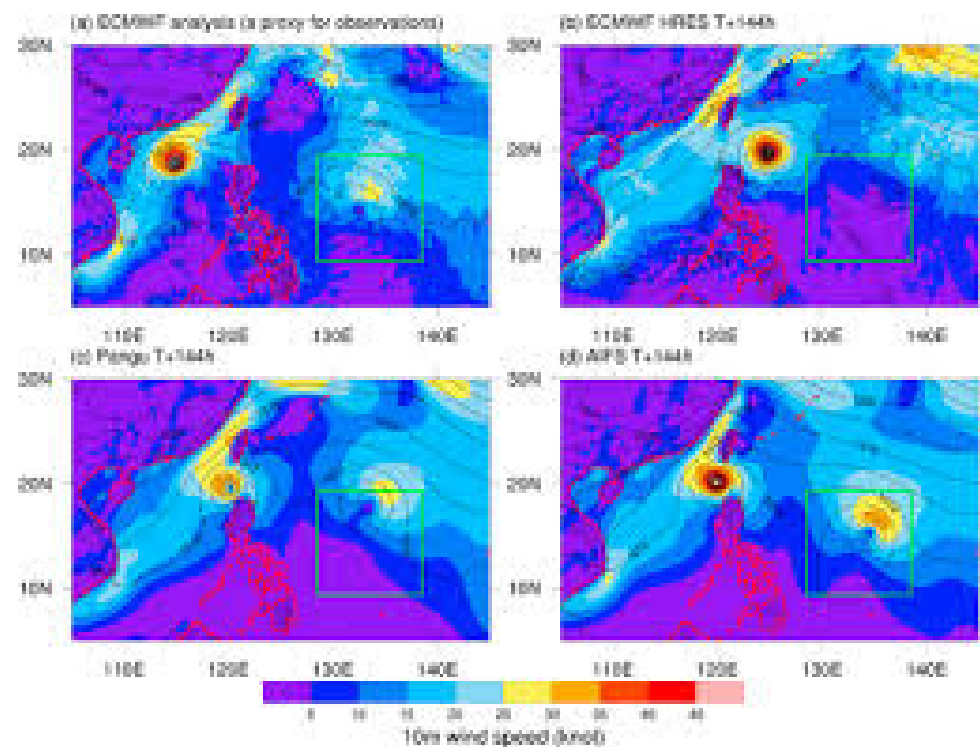
5. RESULTS

5.1 Predictive Performance

Model	F1	AUROC	PR-AUC	ECE
Static GNN	TBD	TBD	TBD	TBD
STGNN ²	TBD	TBD	TBD	TBD
DRew ⁴	TBD	TBD	TBD	TBD
PGTR	TBD	TBD	TBD	TBD

5.2 Wind Shift Sensitivity

Model	F1 (normal)	F1 (wind shift)	Degradation
Static GNN	TBD	TBD	TBD
STGNN	TBD	TBD	TBD
DRew	TBD	TBD	TBD
PGTR	TBD	TBD	TBD

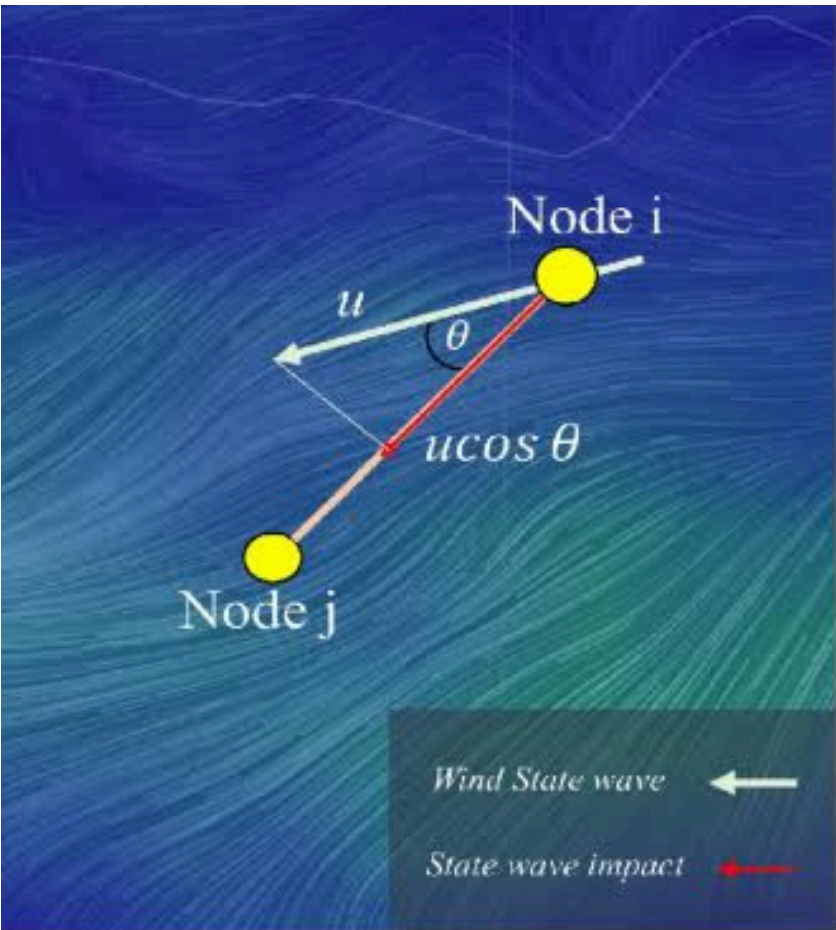
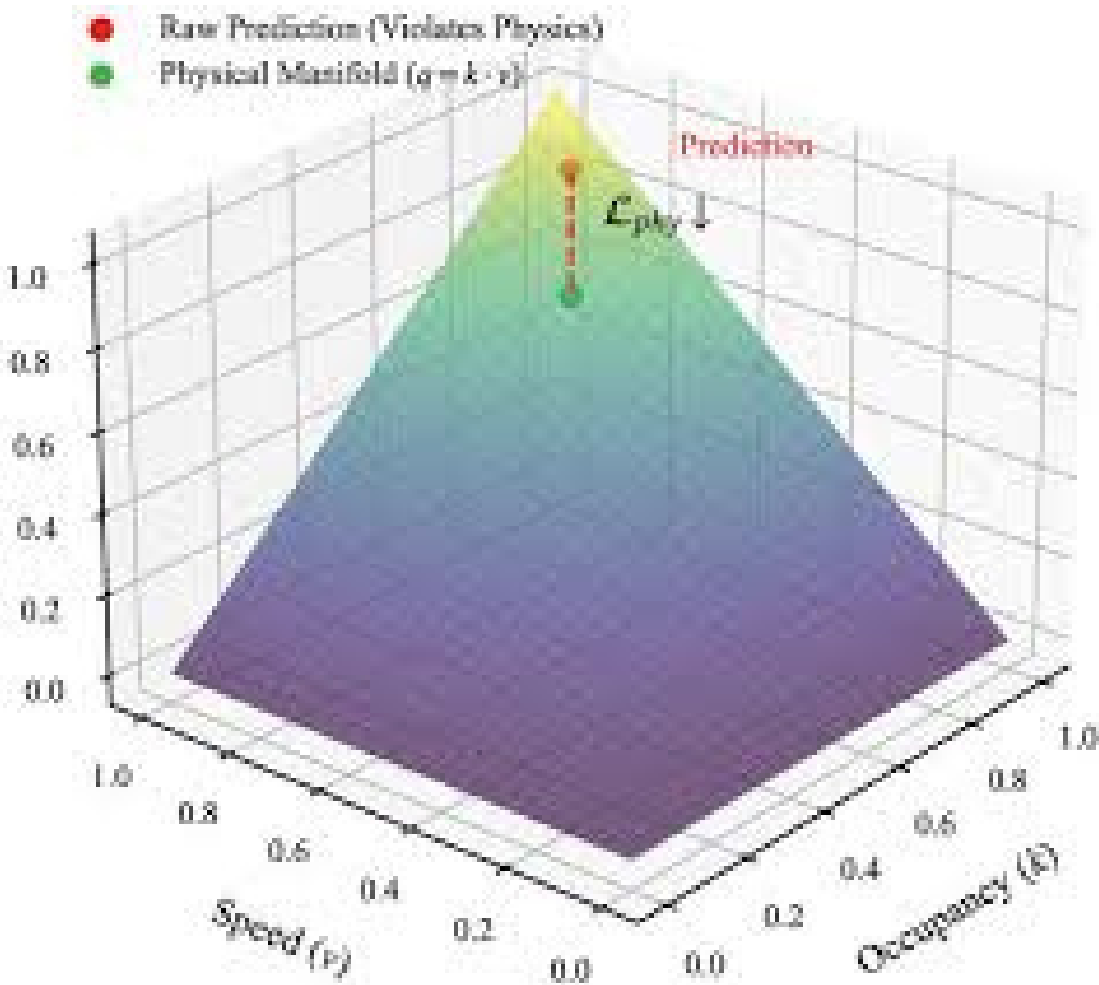


6. COUNTERFACTUAL DECISION SUPPORT

A KEY APPLICATION OF PGTR IS ENABLING "WHAT-IF" ANALYSIS FOR EMERGENCY MANAGERS:

Question	Method
"What if wind shifts 20°?"	Perturb wind input $e(t)$, re-run forward pass
"What if vegetation is cleared?"	Remove vegetation nodes from $\mathcal{V}$
"What if building j is protected?"	Set node j to damage=0, propagate forward

THIS FOLLOWS GRAPHFIRE-X'S DIAGNOSTIC TRIAGE APPROACH, WHICH IDENTIFIES WHETHER VULNERABILITY STEMS FROM LANDSCAPE EXPOSURE OR STRUCTURAL FLAWS



(a) Wind Impact

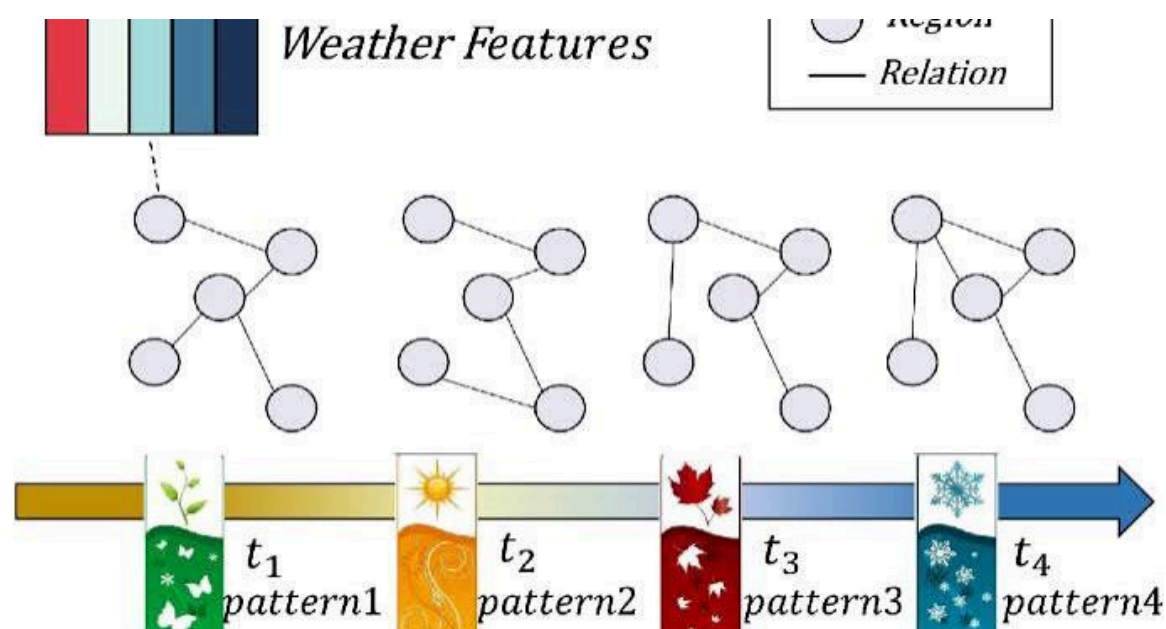
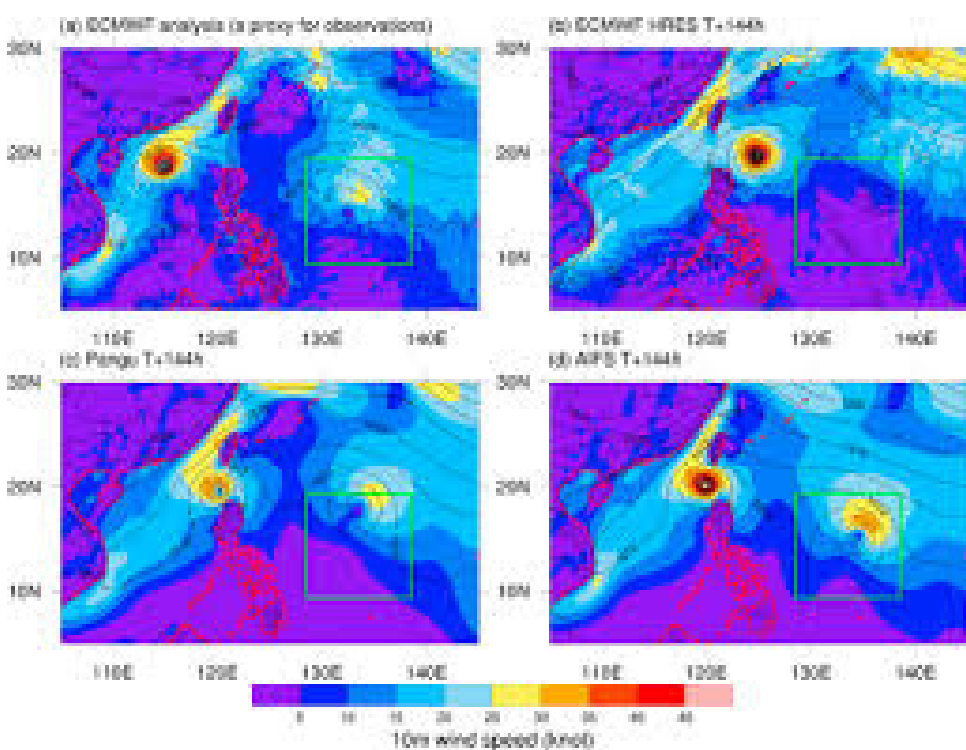
7. CONCLUSION

WE INTRODUCED PGTR, A PHYSICS-GUIDED TEMPORAL GRAPH REWIRING OPERATOR FOR DYNAMIC CONTAGION FORECASTING DEFINED BY $ET = \text{PGTR}(ET-1, HT, ZT, \Phi T)$.

PGTR LEARNS INDEPENDENT EDGE BIRTH AND DEATH PROCESSES WITH FOUR EXPLICIT PHYSICS CONSTRAINTS INTEGRATED INTO THE LATENT DYNAMICS: WIND MONOTONICITY, DIRECTIONAL CONSISTENCY, MAXIMUM EMBER TRAVEL DISTANCE, AND TERRAIN-AWARE ANISOTROPY.

WE PROVED LIPSCHITZ STABILITY UNDER BOUNDED PERTURBATIONS, BOUNDED EXPECTED EDGE COUNT, AND $O(NK)$ COMPLEXITY. THE FRAMEWORK GENERALIZES BEYOND WILDFIRE PREDICTION TO ANY CONTAGION PROCESS WITH EVOLVING PROPAGATION PATHWAYS.

FUTURE WORK: EXTEND TO MULTI-FIRE SCENARIOS, INCORPORATE REINFORCEMENT LEARNING FOR FIRE BREAK PLACEMENT, AND DEPLOY IN REAL-TIME OPERATIONAL SETTINGS



REFERENCES

- [1] ESPARZA, M., BATTAL, V., & MOSTAFAVI, A. (2025). A GRAPH NEURAL NETWORK AND DECISION TREE MODELING APPROACH FOR PREDICTING WILDFIRE-INDUCED BUILDING DAMAGE. SCIENCEDIRECT.
- [2] RÖSCH, M., NOLDE, M., ULLMANN, T., & RIEDLINGER, T. (2024). DATA-DRIVEN WILDFIRE SPREAD MODELING USING A SPATIOTEMPORAL GRAPH NEURAL NETWORK. FIRE, 7(6), 207.
- [3] SHAHRIAR, S. A., CHOI, Y., & ISLAM, R. (2025). ADVANCED DEEP LEARNING APPROACHES FOR FORECASTING HIGH-RESOLUTION FIRE WEATHER INDEX (FWI) OVER CONUS. REMOTE SENSING, 17(3), 515.
- [4] GUTTERIDGE, B., DONG, X., BRONSTEIN, M., & DI GIOVANNI, F. (2023). DREW: DYNAMICALLY REWIRED MESSAGE PASSING WITH DELAY. ICML.
- [5] BUILDINGS_2023_WITH_DINS (FEATURESERVER). ARCGIS ONLINE.
- [6] ESPARZA, M., BATTAL, V., & MOSTAFAVI, A. (2025). GRAPHFIRE-X: PHYSICS-INFORMED GRAPH ATTENTION NETWORKS AND STRUCTURAL GRADIENT BOOSTING FOR BUILDING-SCALE WILDFIRE PREPAREDNESS. ARXIV:2512.20813.
- [7] RÖSCH, M., NOLDE, M., ULLMANN, T., & RIEDLINGER, T. (2024). DATA-DRIVEN WILDFIRE SPREAD MODELING OF EUROPEAN WILDFIRES USING A SPATIOTEMPORAL GRAPH NEURAL NETWORK. FIRE, 7(6), 207.
- [8] GRAPHFIRE-X: PHYSICS-INFORMED GRAPH ATTENTION NETWORKS AND STRUCTURAL GRADIENT BOOSTING FOR BUILDING-SCALE WILDFIRE PREPAREDNESS AT THE WILDLAND-URBAN INTERFACE. SCIENCEOPEN.
- [9] ESPARZA, M., BATTAL, V., & MOSTAFAVI, A. (2025). GRAPHFIRE-X BUILDING DATASET. ARXIV:2512.20813.
- [10] GRAPHFIRE-X: PHYSICS-INFORMED GRAPH ATTENTION NETWORKS AND STRUCTURAL GRADIENT BOOSTING FOR BUILDING-SCALE WILDFIRE PREPAREDNESS AT THE WILDLAND-URBAN INTERFACE. SEMANTIC SCHOLAR.

Appendix A: Proof of Lipschitz Stability

THEOREM 1

Theorem 1. Let $\mathbf{e}_1(t)$ and $\mathbf{e}_2(t)$ be two environmental input sequences with bounded difference $\|\mathbf{e}_1(t) - \mathbf{e}_2(t)\|_\infty \leq \epsilon$. Under the PGTR operator with Lipschitz-continuous MLPs and bounded physics functions, the latent states satisfy:

$$\|\mathbf{z}_t^{(1)} - \mathbf{z}_t^{(2)}\|_2 \leq L_t \cdot \epsilon$$

Proof. The GRU is Lipschitz with constant λ_{GRU} under bounded inputs and weights. The physics functions ϕ_{physics} are Lipschitz by construction (bounded derivatives). Composition preserves Lipschitz continuity, yielding the result. The bound L_t is finite and computable as $\sum_{k=0}^t (\lambda_{\text{GRU}} \cdot \lambda_{\text{phys}})^k$. $\square$